Ali Haidar

AI/ML & Robotics Engineer | Computer Vision · Deep Learning · Applied ML in Automation

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SUMMARY

AI/ML & Robotics Engineer with an MSc (Honors) in Robotics and Artificial Intelligence from ITMO University, specializing in applied generative AI, production ML systems, and deep learning-based computer vision. Built an agentic RAG chatbot backend for bilingual restaurant and gym clients, with logging and observability via Arize Phoenix. At IROS-System, developed a cost-prediction regression model and a cosine-similarity-based part-matching system from BOM and assembly data, alongside CAD automation and FEA validation for industrial automation. Graduate research focused on real-time visual object tracking using deep learning techniques, complemented by a mechatronics and robotics background spanning motion control, kinematics, and mechatronic system design.

EXPERIENCE

AI/ML & CAD Engineer — Industrial Automation — IROS-System — **Jul 2023 – Present**

Saint Petersburg, Russia

- Built a cost-prediction regression model and a cosine-similarity-based part-matching system (Python, pandas, scikit-learn) from BOM and assembly data, enabling earlier cost estimates and reducing part duplication.
- Designed 3D part models, assemblies, and technical drawings, and built rule-based automation scripts (iLogic, Inventor API) to auto-generate design variants and drawings, cutting modeling time and reducing manual errors.
- Performed stress calculations and FEA validation across all design outputs, maintaining 100% compliance with regulatory standards.

Freelance AI/ML Engineer — Self-employed — **Aug 2025 – Present**

Remote

- Designed and shipped an AI agent for a restaurant chain's RAG chatbot backend (bilingual RU/EN menu + FAQ), using FastAPI, LangGraph, and Weaviate Cloud for retrieval.
- Implemented logging, monitoring, and observability with Arize Phoenix to track and debug agent behavior in production.
- Delivered an AI agent-based chatbot engagement for a gym client.

Research Assistant & Robotics Engineer — Higher Institute for Applied Sciences and Technology (HIAST) — **Oct 2018 – Oct 2022**

Damascus, Syria

- Supervised 4th- and 5th-year capstone projects in robotics and mechatronics, including a Knee CPM machine, a propulsive bionic foot, and a laser engraving machine.
- Taught Industrial Drawing and CAD to 100+ students; led advanced modules in Robotics, Industrial Automation, and CAM.
- Delivered practical training in Ansys simulation and PLC coding, preparing students for industrial manufacturing roles.

Mechatronics Engineer, Automated Systems (Part-time) — Syriatel Mobile Telecom and Milkman — **Jun 2018 – Jun 2020**

Damascus, Syria

- Designed and implemented a custom 500×500×500mm Cartesian 3D printer, managing the full lifecycle from mechanical design to firmware integration.

- Developed a high-precision volumetric filling machine, a cup feeder machine, and an end-line picker machine for high-viscosity, food-grade production lines.
- Managed development of a Building Management System (BMS), improving facility automation and energy efficiency.

EDUCATION

MSc (Honors), Robotics and Artificial Intelligence — ITMO University **Oct 2024 – Jul 2025**

Saint Petersburg, Russia

- Thesis: "Real-Time Visual Object Tracking Methods with Deep Learning Techniques."
- Coursework: Machine Learning in Robotics, Introduction to ML Tools and Applied AI in Science and Business, Data Preprocessing and Big Data Storage Technologies, Cyber-Physical Systems and Technologies, Robot Programming, Robot Modeling and Motion Control, Trajectory Planning and Control, Simulation of Robotic Systems, Design of Mechatronic Systems.
- Research focus: implementing deep learning algorithms for visual perception in dynamic environments.

BEng, Mechatronics Engineering — Higher Institute for Applied Sciences and Technology (HIAST) **Oct 2013 – Oct 2018**

Damascus, Syria

- Graduation project: designed and manufactured a Delta Linear Robot with 0.1mm positioning accuracy for laser drawing and 3D printing applications.
- 4th-year project: designed an open-wheel electric vehicle (Go-Kart).

SKILLS

AI/ML & Data Science: Machine Learning, Deep Learning, Computer Vision, Artificial Neural Networks, Image Processing, Reinforcement Learning, TensorFlow, Python (pandas, scikit-learn)

Generative AI & LLMs (applied): Retrieval-Augmented Generation (RAG), AI Agents / LangGraph, Large Language Models (LLMs), Generative AI

Robotics: Robot Operating System (ROS), Motion Control & Kinematics, Trajectory Planning and Control, Mechatronic System Design

Engineering & CAD: Autodesk Inventor, Autodesk Vault, AutoCAD, SOLIDWORKS, Computer-Aided Design (CAD), Finite Element Analysis (FEA), CAD Standards & Bill of Materials, Technical Drawing, CNC Manufacturing, 3D Printing, MATLAB

Tools & Infrastructure: FastAPI, Weaviate, Docker, Git, Programmable Logic Controller (PLC), iLogic

CERTIFICATIONS

- Generative AI with Large Language Models — DeepLearning.AI (Jul 2026)
- Retrieval Augmented Generation (RAG) — DeepLearning.AI (Jul 2026)
- AI Agents in LangGraph — DeepLearning.AI (Jun 2026)
- Functions, Tools and Agents with LangChain — DeepLearning.AI (Jun 2026)
- LangChain for LLM Application Development — DeepLearning.AI ([NEED DATE])
- LangChain Chat with Your Data — DeepLearning.AI ([NEED DATE])
- Knowledge Graphs for RAG — DeepLearning.AI (Jun 2026)
- Robotics and Artificial Intelligence (Sber × ITMO) — Sberbank-Technology (Jul 2025)
- Docker Certified Associate Exam Prep Specialization — LearnKartS (Jul 2026)

LANGUAGES

Arabic (Native) · English (B2) · Russian (B1)